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Assignment 13

Interim Documentation

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Risk Register

Prepared for Saint Cedric Ministry of Health regulatory consultation, via Dr De Fretias, Mercer
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Purpose

This register identifies 10 named risks spanning the AI-technical, operational, ethical, and equity dimensions of a phased AI-assisted triage rollout, modelled on the Phase 1 (digital capture) / Phase 2 (rule-based and ML classifier) approach proposed for Mercer General ED. Each risk includes likelihood, impact, a proposed mitigation, and a measurable signal that the mitigation is actually working — not just present on paper.

Dr De Fretias's instruction: ***do not flinch***. Where a mitigation is weak or unproven, that is stated plainly rather than smoothed over.

Risk Register Table

#	Risk	Category	L	I	Mitigation	Signal of Success
1	Distribution shift / single-site overfitting. A model trained on Mercer's patient population performs poorly when deployed at a different hospital with a different demographic and disease mix.	AI-technical	H	H	Mandate site-specific external validation before any new deployment; no “train once, deploy everywhere” rollout.	AUC/sensitivity tracked per site quarterly; deployment paused if performance drops below a pre-agreed local threshold.
2	Algorithmic bias from non-representative training data. No Caribbean ED	AI-technical	H	H	Phase 1 deliverable is exclusively Caribbean-sourced data;	Published demographic audit comparing training set

	triage dataset currently exists; bootstrapping from non-Caribbean data risks encoding assumptions that don't hold locally.				no transfer learning from non-Caribbean models without full local revalidation.	composition against the ED's actual catchment population.
3	Silent failure — no uncertainty signal on ESI suggestions. A classifier outputs a triage suggestion with no indication of how confident or borderline it is.	AI-technical	M	H	Every suggestion must surface the triggering rule or confidence score alongside the ESI level — never a bare number.	100% of system-generated suggestions are paired with an explainable trigger or confidence indicator (audited at code review).
4	Alert fatigue from over-triage. High false-positive volume	Operational	H	H	Conservative initial alert thresholds; minimum locally-validated positive	Alerts-per-true-positive ratio tracked monthly; target materially better than the 109:1 ratio

	causes nurses to start ignoring or dismissing alerts reflexively, including true positives.				predictive value required before go-live; nurse-tunable thresholds post-deployment.	documented in the Epic Sepsis Model case (see Part 2).
5	Workflow friction is causing data-entry shortcuts. Nurses copy-paste from the previous patient or skip fields if the digital tool takes longer than the paper form.	Operational	H	M	Phase 1 fields are capped at what the existing paper form already captures; usability testing confirms ≤ 5 -minute completion time.	Field completion audit; target $>95\%$ of entries are genuine (non-default, non-duplicated) values.
6	Automation bias. Nurses defer to the system's suggested ESI level even	Ethical	M	H	System is framed explicitly as decision support, never decision	Override rate tracked over time. Caveat: a near-zero override rate is NOT success — it may indicate

	when it contradicts their own clinical judgment, particularly under time pressure.				maker; any override requires the nurse to document a one-line reason.	automation bias. Needs a paired qualitative audit.
7	Informed consent gap. Patients are triaged, and their data feeds a digital/AI system without being meaningfully informed this is happening.	Ethical	M	M	Visible signage and a short verbal consent disclosure at registration; an opt-out pathway that does not delay or downgrade care.	Spot-check audit of registration encounters confirming disclosure was given.
8	Deskilling — erosion of independent triage judgment over time. Long-term reliance on	Ethical	L- M	H	Periodic “blind” triage audits where nurses triage without system input, to	Blind-audit ESI accuracy held stable over time relative to the pre-deployment baseline.

	system suggestions could degrade nurses' own unaided triage accuracy.				confirm skill retention against baseline.	
9	Training data demographic skew. Mercer's data is from one urban Caribbean ED; a model trained on it may not generalise to rural EDs or other island contexts.	Equity	H	H	Deployment scope explicitly limited to validated sites; any new site requires local revalidation before go-live — mirroring Epic's own corrective step (see Part 2).	Site-specific validation report is a hard gate before any new deployment — not a recommendation, a requirement.
10	Differential reliability across patient subgroups. Aggregate accuracy can mask poor performance for	Equity	M	H	Performance is reported by subgroup (age band, presentation type), never aggregate-only.	Subgroup-level sensitivity/specificity is published alongside overall figures in every validation report.

	paediatric patients, elderly patients with atypical presentations, or non-standard dialect free-text capture.					
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Risk Memo

1. Alert Fatigue (Risk 4)

If the system cries wolf too often, nurses stop listening — including when it's right. This is not a hypothetical: it is the single most consistent failure mode found in the deployed clinical AI literature reviewed this week, most starkly in the Epic Sepsis Model, where clinicians had to investigate 109 alerts to find one genuine sepsis case. A Mercer system that floods triage nurses with low-value flags will be tuned out within days, regardless of how good the underlying logic is — and once nurses learn to ignore it, even a correct alert is at risk of being dismissed.

2. Distribution Shift / Single-Site Overfitting (Risk 1)

A model is only as good as the population it was trained on. Mercer's eventual dataset will reflect one hospital, in one part of the Caribbean, over one stretch of time. If that model is later assumed to work identically at a different hospital — different demographics, different disease burden, different staffing — it can quietly fail in ways nobody notices until patients are harmed. This is exactly what happened with the Epic Sepsis Model: a tool validated in one context was deployed at hundreds of hospitals with wildly different patient populations, and its real-world performance fell far short of what internal testing had suggested.

3. Training Data Demographic Skew (Risk 9)

This is the equity risk underlying the entire project's justification. The whole premise of the Phase 1 proposal is that AI triage tools built on North American and European data don't transfer to the Caribbean — but that same logic applies within the Caribbean too. A model trained only on Mercer's urban catchment will not automatically work for a rural clinic in a different territory with a different patient mix. Solving the “Caribbean data gap” with a single urban dataset risks just relocating the same generalisability problem to a smaller scale, rather than actually solving it.

AI-Harm Case Study: The Epic Sepsis Model

What Happened

The Epic Sepsis Model (ESM) is a proprietary, AI-driven early-warning tool embedded in the Epic electronic health record system, designed to alert clinicians when a hospitalised patient may be developing sepsis. It was deployed at hundreds of US hospitals on the strength of internal validation figures Epic had reported, without mandatory independent external validation before clinical use.

In June 2021, Wong et al. (2021) published an external validation of the ESM in JAMA Internal Medicine, examining 38,455 hospitalisations at Michigan Medicine between December 2018 and October 2019. The results fell sharply short of Epic's internal claims: a sensitivity of only 33% (the model missed two-thirds of actual sepsis cases), a positive predictive value of just 12%, and an area under the curve (AUC) of 0.63 — well below the 0.76–0.83 range Epic had cited internally. The clinical burden was severe: clinicians needed to investigate roughly 109 alerts to find one genuine sepsis case. A second, independent study across two county-hospital emergency departments later replicated this failure pattern, finding the model missed sepsis in over half of encounters and offered little to no time advantage over standard clinical recognition.

The Root Cause. Why did it happen?

The core failure was not a single bug but a generalisability gap baked into the model's design and deployment process. The ESM was developed and tuned on data from one set of hospital systems, then deployed broadly under the assumption that its performance would transfer

unchanged to any hospital running Epic — different patient populations, different disease prevalence, different documentation habits and all. STAT News investigations later surfaced an additional design flaw: the model used whether a patient had already received antibiotics as a predictive input, meaning it was partly learning to detect sepsis that clinicians had already caught and treated, rather than predicting it ahead of time. This produced a model that looked statistically respectable in aggregate but performed poorly at the one task that actually mattered clinically: catching sepsis before a human would.

Critically, the system was deployed at scale without a mandatory requirement for local validation. Hospitals were trusting Epic's internal performance figures rather than testing the tool against their own patient population before going live.

What the System Failed to Anticipate

The model's developers failed to anticipate that statistical performance on a training population does not guarantee performance on a different population — the exact distribution-shift risk named as Risk 1 in this project's register. They also failed to anticipate the operational consequence of a high false-positive rate: a 109:1 alert-to-true-positive ratio is not a minor inconvenience; it is a near-guarantee of alert fatigue, where clinicians begin reflexively dismissing alerts, including correct ones.

What Would Have Caught It

Mandatory external validation at each deploying hospital, before clinical go-live, would have surfaced this gap immediately — Michigan Medicine's own data showed the problem the moment someone looked. A requirement to report sensitivity, specificity, and PPV by deploying the site rather than relying on a single internal benchmark would have made the risk visible long before hundreds of hospitals had already adopted the tool. This is precisely the mitigation proposed for Risks 1 and 9 in this register: no deployment to a new site without local revalidation first.

Relevance to This Project

This case is a close analogue to the Phase 2 ambition of this proposal — an early-warning, decision-support tool sitting inside a clinical workflow, prioritising patients by risk. The lesson transfers directly: any classifier trained on Mercer's eventual dataset must be locally validated before being assumed to work elsewhere, and alert volume must be tuned conservatively from the start rather than corrected after clinicians have already learned to ignore it.

References

- Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., Pestruie, J., Phillips, M., Konye, J., Penzoza, C., Ghous, M., & Singh, K. (2021). External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. *JAMA Internal Medicine*, 181(8), 1065–1070. <https://doi.org/10.1001/jamainternmed.2021.2626>
- Habib, A. R., Lin, A. L., & Grant, R. W. (2021). The Epic Sepsis Model Falls Short — The Importance of External Validation. *JAMA Internal Medicine*, 181(8), 1040–1041. <https://doi.org/10.1001/jamainternmed.2021.3333>

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol